

Quantum Mechanics

Review: Electromagnetic wave

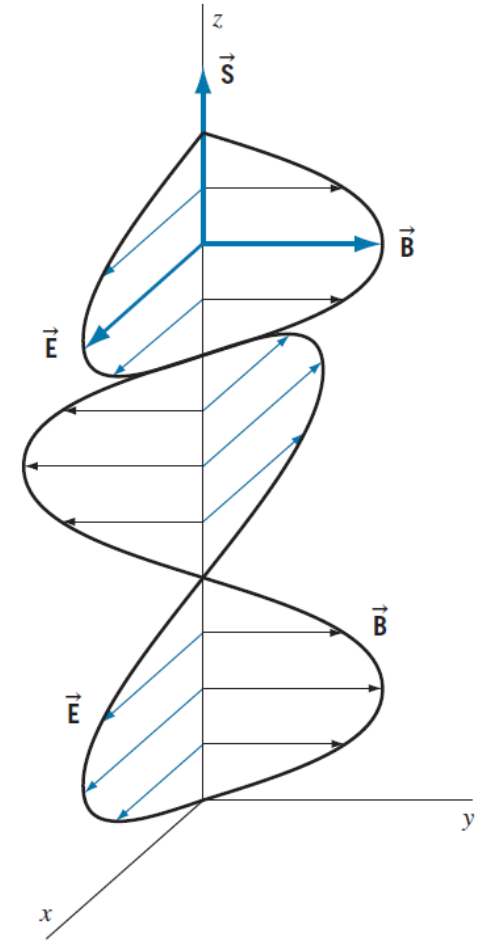
$$\nabla^2 \mathbf{E} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{E}}{\partial t^2} \quad \nabla^2 \mathbf{B} = \mu_0 \epsilon_0 \frac{\partial^2 \mathbf{B}}{\partial t^2}$$

$$\vec{\mathbf{E}} = \vec{\mathbf{E}}_0 \sin(kz - \omega t), \quad \vec{\mathbf{B}} = \vec{\mathbf{B}}_0 \sin(kz - \omega t) \quad B_0 = \frac{E_0}{c}$$

Poynting vector :

$$\vec{\mathbf{S}} = \frac{1}{\mu_0} \vec{\mathbf{E}} \times \vec{\mathbf{B}}$$

$$\begin{aligned} \vec{\mathbf{S}} &= \frac{1}{\mu_0} E_0 B_0 \sin^2(kz - \omega t) \hat{\mathbf{k}} \\ &= \frac{1}{\mu_0 c} E_0^2 \sin^2(kz - \omega t) \hat{\mathbf{k}} \end{aligned}$$



The Poynting vector has dimensions of power (energy per unit time) per unit area

Review: Electromagnetic wave

The power propagates along z direction is

$$P = \frac{1}{\mu_0 c} E_0^2 A \sin^2(kz - \omega t)$$

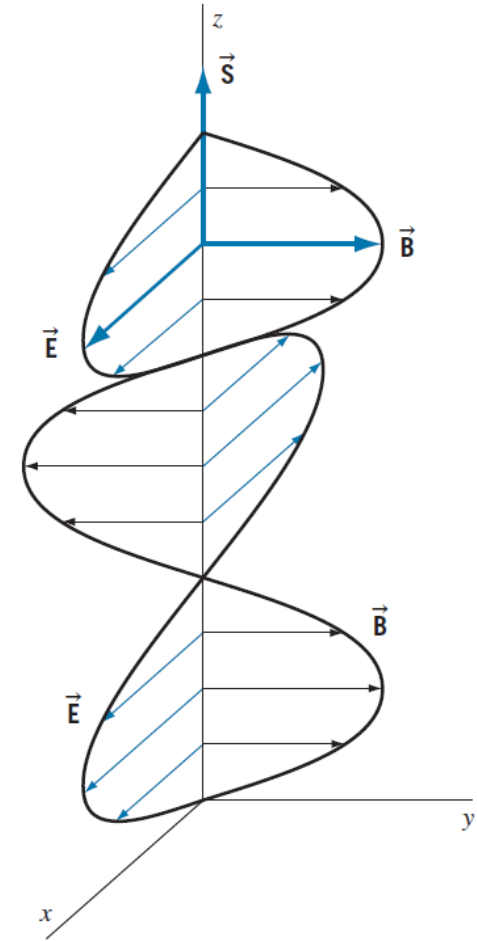
Energy carried through wave-front

Important NOTE:

- The power fluctuates with time with a high frequency $2(\omega/2\pi) \approx 10^{15}$

We observe an average value $P_{\text{av}} = \frac{1}{T} \int_0^T P dt$

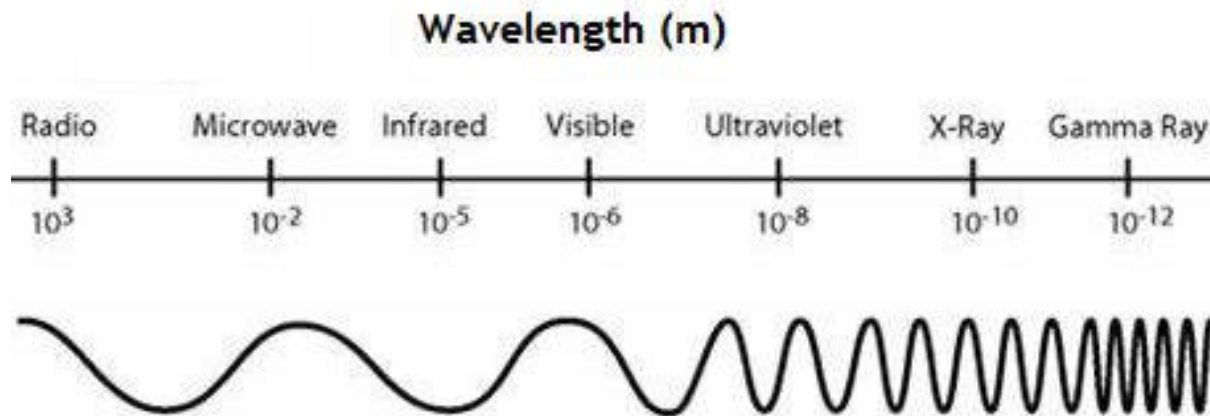
- The intensity (the average power per unit area) is proportional to E_0^2 . This is a general property of waves: *the intensity is proportional to the square of the amplitude.*



Review: Electromagnetic wave

Velocity of the electromagnetic radiation is constant in free space

$$c = \frac{1}{\sqrt{\mu_0 \epsilon_0}} = \lambda \cdot \nu \cong 3 \times 10^8 \text{ m/s}$$



Using the wave nature of light was difficult to explain the nature of the black body radiation spectrum !!!

Black-body radiation

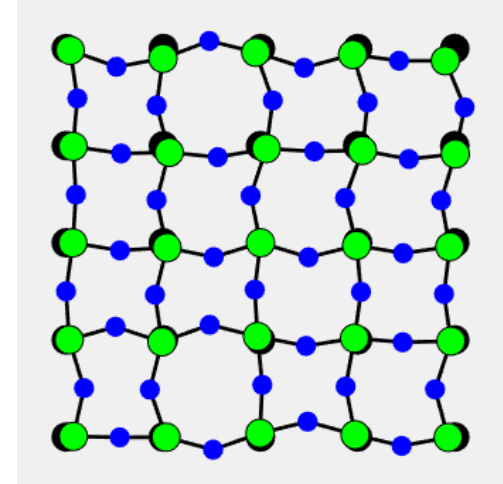
What is black-body ??

It is a substance which does not reflect electromagnetic radiation !!

After absorption, black body radiates (How?)

Many experiments were done in the late 19th century to study the wavelength spectrum of thermal radiation. These experiments, as we shall see, gave results that totally disagreed with the predictions of the classical theories of thermodynamics and electromagnetism.

The goal of these experiments was to measure the intensity of the radiation emitted by the object as a function of wavelength.



Black-body radiation

Observation :

The total intensity radiated over all wavelengths (that is, the area under each curve) increases as the temperature is increased.

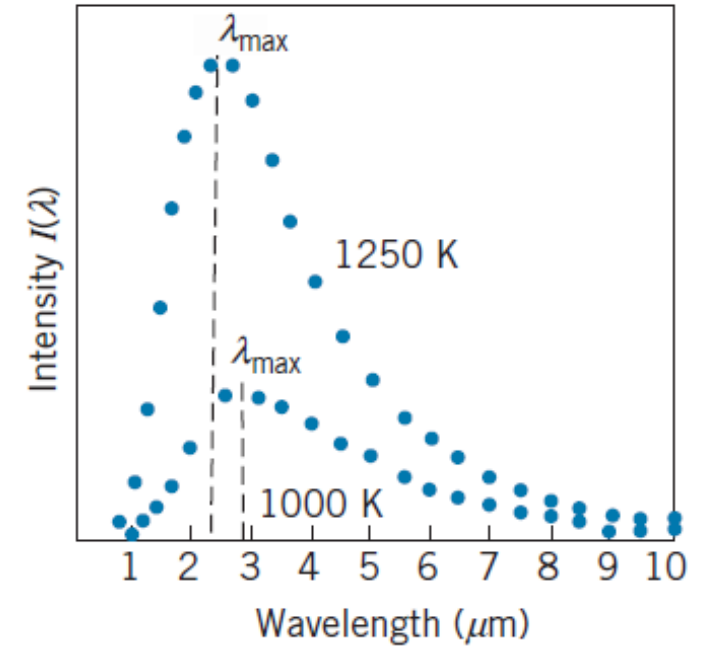
From careful measurement, it was found that the total intensity increases as the fourth power of the absolute or kelvin temperature:

$$I = \sigma T^4 \quad \text{Stefan's law}$$

$\sigma = 5.67037 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$ is called the *Stefan-Boltzmann constant*.

The wavelength λ_{max} at which the emitted intensity reaches its maximum value decreases as the temperature is increased, in inverse proportion to the temperature: $\lambda_{\text{max}} \propto 1/T$. The proportionality constant is called Wien's constant.

$$\lambda_{\text{max}} T = 2.8978 \times 10^{-3} \text{ m} \cdot \text{K} \quad \text{Wien's displacement law}$$

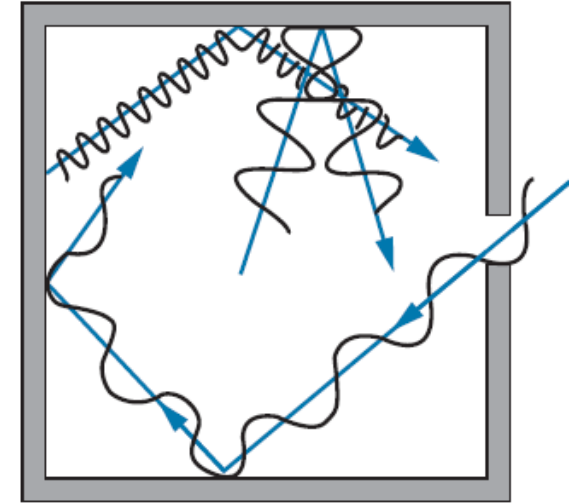


Black-body radiation

Theoretical analysis:

Black-body constriction : Small hole in the surface of a metal sphere

- All the radiation reach at the hole will be absorb.
- Radiation continuously reflect back and forth from the shiny surface and finally forms standing wave and absorbs



Black-body radiation

Theoretical analysis:

The energy density (energy per unit volume) per unit wavelength interval $u(\lambda)$

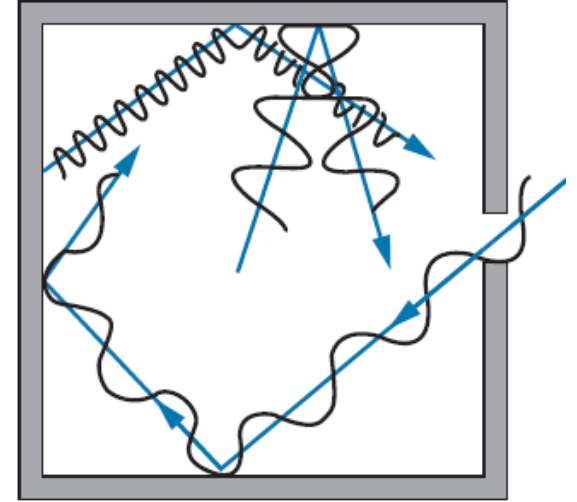
The energy density of the electromagnetic radiation with wavelengths between λ and $\lambda + d\lambda$ in a small volume element is $u(\lambda)d\lambda$

Half of the total radiation is moving toward the hole at velocity of magnitude c but directed over a range of angles. Averaging over this range of angles to evaluate the energy flowing perpendicular to the surface of the hole introduces another factor of $\frac{1}{2}$, so the contribution of the radiation in this small wavelength interval to the intensity passing through the hole is:

$$I(\lambda) = \frac{c}{4}u(\lambda)$$

$$I(\lambda_1:\lambda_2) = \int_{\lambda_1}^{\lambda_2} I(\lambda) d\lambda$$

$$I = \int_0^{\infty} I(\lambda) d\lambda$$



Black-body radiation

Theoretical analysis:

$$I(\lambda) = \frac{c}{4} u(\lambda)$$

If λ_1 & λ_2 are two possible standing wave then

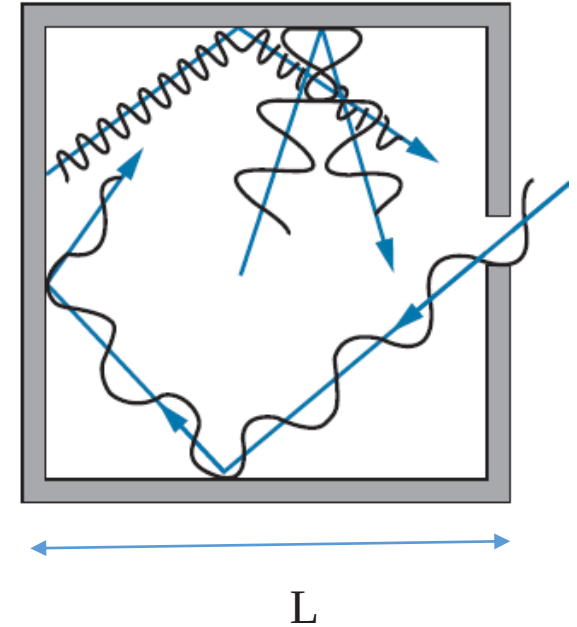
$$\begin{aligned} L = \frac{n_1 \lambda_1}{2} &\Rightarrow n_1 = \frac{2L}{\lambda_1} \\ L = \frac{n_2 \lambda_2}{2} &\Rightarrow n_2 = \frac{2L}{\lambda_2} \\ n &= \frac{2L}{\lambda} \end{aligned} \quad n_2 - n_1 = 2L \left(\frac{1}{\lambda_2} - \frac{1}{\lambda_1} \right)$$

Number of wave in the box whose wave length is in between λ and $\lambda + d\lambda$ will be

$$N(\lambda)d\lambda = \left| \frac{dn}{d\lambda} \right| d\lambda = \frac{2L}{\lambda^2} d\lambda$$

In 3d

$$N(\lambda)d\lambda = \frac{8\pi V}{\lambda^4} d\lambda$$



Black-body radiation

Theoretical analysis:

$$N(\lambda)d\lambda = \frac{8\pi V}{\lambda^4} d\lambda$$

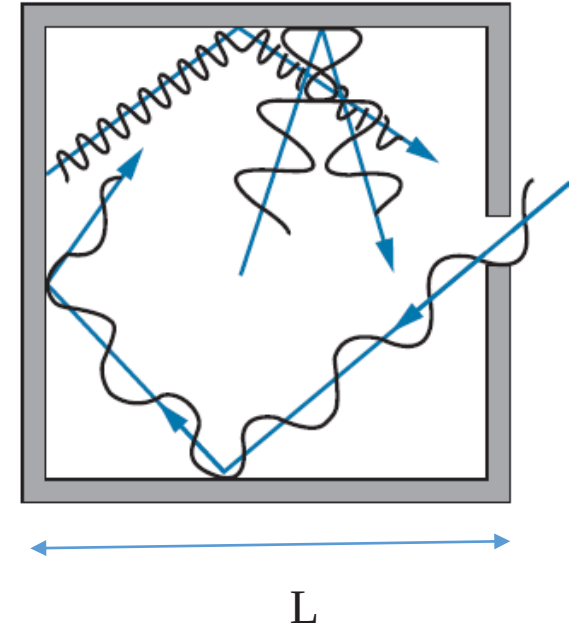
The energy distribution over the oscillator follows the Maxwell-Boltzmann distribution

$$N(E) = \frac{N}{kT} e^{-E/kT}$$

The average energy of an oscillator :

$$E_{\text{av}} = \frac{1}{N} \int_0^{\infty} E N(E) dE = \frac{1}{kT} \int_0^{\infty} E e^{-E/kT} dE$$

$$E_{\text{av}} = kT.$$



Black-body radiation

Theoretical analysis:

$$N(\lambda)d\lambda = \frac{8\pi V}{\lambda^4} d\lambda$$

Each of these wave associate with the energy kT

The energy inside the box due to those wave have wave length is in between λ and $\lambda + d\lambda$ is

$$N(\lambda)d\lambda \times kT = \frac{8\pi V kT}{\lambda^4} d\lambda$$

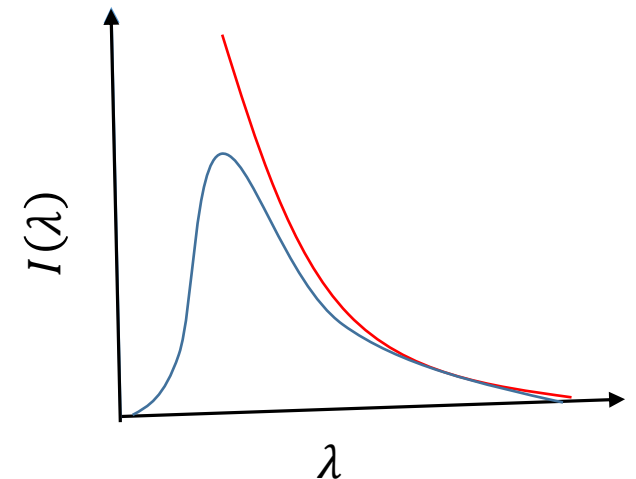
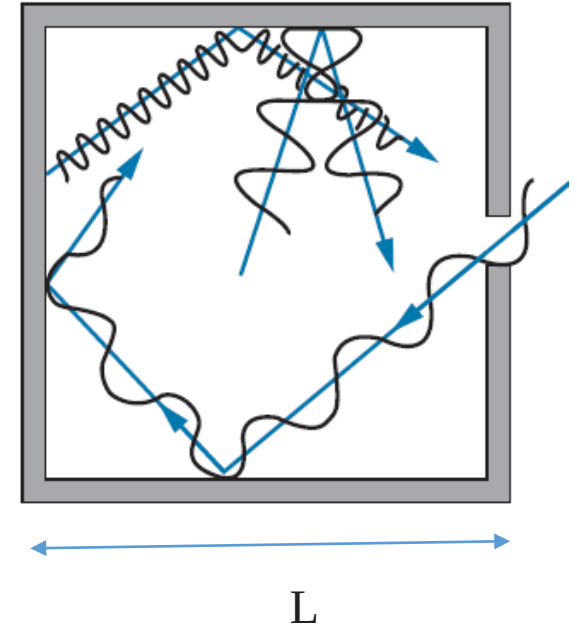
The energy density inside the box withing the wavelength interval λ and $\lambda + d\lambda$ is

$$u(\lambda)d\lambda = \frac{N(\lambda)d\lambda kT}{V} = \frac{8\pi kT}{\lambda^4} d\lambda$$

$$I(\lambda)d\lambda = \frac{c}{4} u(\lambda)d\lambda = \frac{2\pi c kT}{\lambda^4} d\lambda$$

$$I(\lambda)d\lambda = \frac{2\pi c kT}{\lambda^4} d\lambda$$

The Rayleigh-Jeans formula



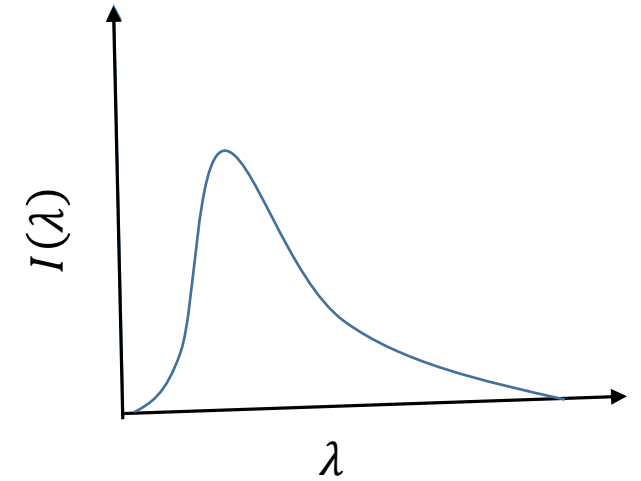
Black-body radiation

Theoretical analysis:

Plank's Theory :

Thought something is wrong in calculating the number of wave for high frequency (low λ) oscillators.

Tries to develop theory for which $u(\lambda) \rightarrow 0$ as $\lambda \rightarrow 0$



Planck suggestion:

An oscillating atom can absorb or emit energy only in discrete bundles.

Each oscillator can emit or absorb energy only in quantities that are integer multiples of a certain basic quantity of energy ε

$$E_n = n\varepsilon, \quad n = 1, 2, 3, \dots \quad \varepsilon = hf$$

Energy of an oscillator at a certain wavelength or frequency is a discrete variable

The number of N_n oscillators with energy E_n can be obtained from the discrete distribution

$$N_n = N(1 - e^{-\varepsilon/kT})e^{-n\varepsilon/kT}$$

Black-body radiation

Theoretical analysis:

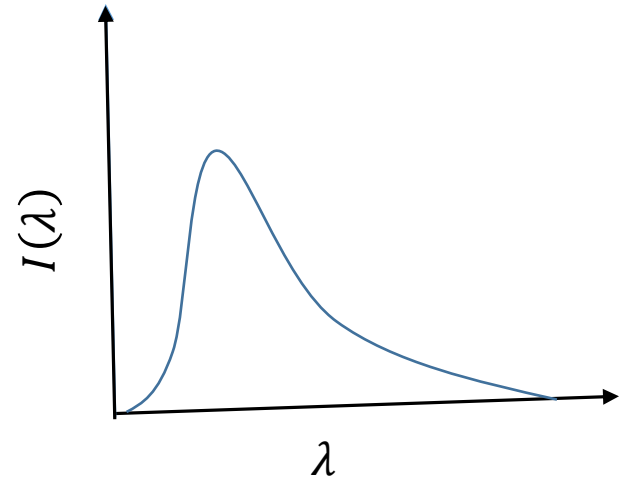
Plank's Theory :

$$N_n = N(1 - e^{-\varepsilon/kT})e^{-n\varepsilon/kT}$$

$$\sum_{n=0}^{\infty} N_n = N$$

$$E_{\text{av}} = \frac{1}{N} \sum_{n=0}^{\infty} N_n E_n$$

$$= (1 - e^{-\varepsilon/kT}) \sum_{n=0}^{\infty} (n\varepsilon) e^{-n\varepsilon/kT}$$



Black-body radiation

Theoretical analysis:

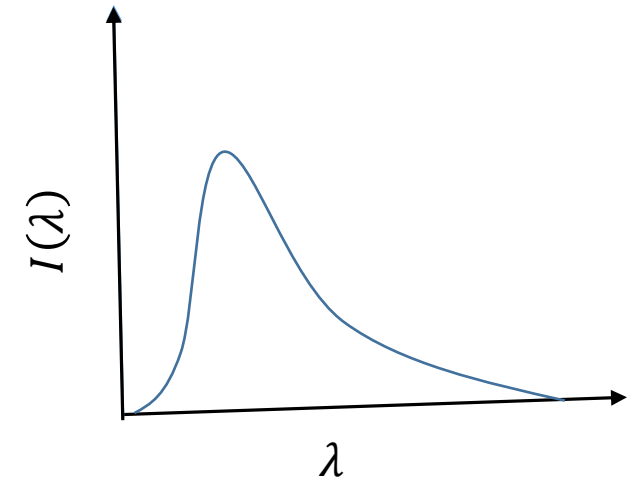
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Black-body radiation

Theoretical analysis:

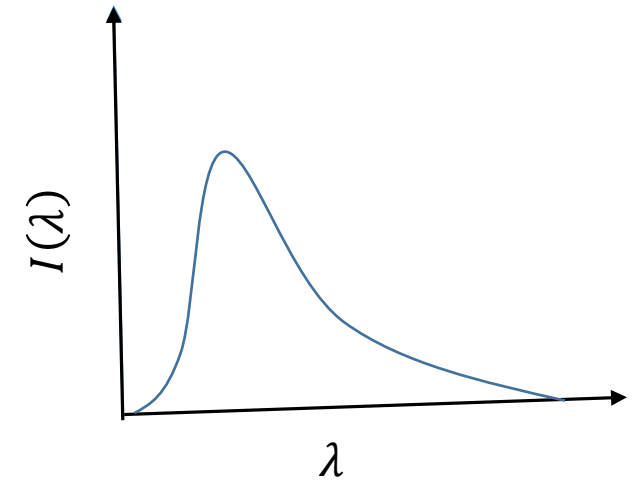
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Black-body radiation

Theoretical analysis:

Plank's Theory :

$$N_n = N(1 - e^{-\varepsilon/kT})e^{-n\varepsilon/kT}$$

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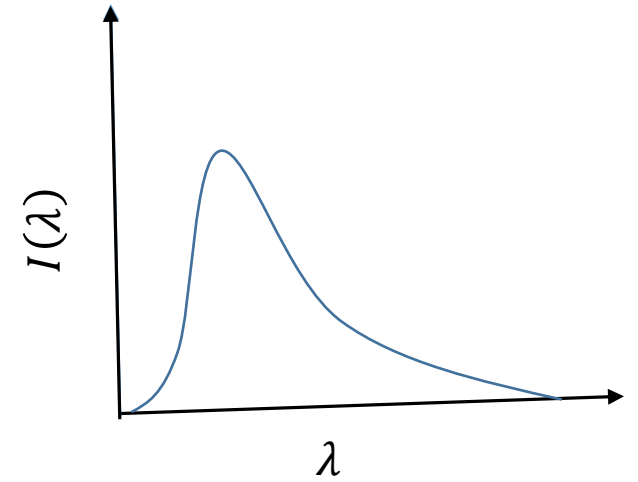
$$E_{\text{av}} = \frac{1}{N} \sum_{n=0}^{\infty} N_n E_n$$

$$= (1 - e^{-\varepsilon/kT}) \sum_{n=0}^{\infty} (n\varepsilon) e^{-n\varepsilon/kT}$$

$$= \frac{\varepsilon e^{-\varepsilon/kT}}{1 - e^{-\varepsilon/kT}}$$

$$I(\lambda) = \frac{c}{4} \left(\frac{8\pi}{\lambda^4} \right) E_{\text{av}}$$
$$= \frac{c}{4} \left(\frac{8\pi}{\lambda^4} \right) \left[\frac{hc/\lambda}{e^{hc/\lambda kT} - 1} \right]$$

$$= \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$



Black-body radiation

Theoretical analysis:

Plank's Theory :

$$I(\lambda) = \frac{2\pi hc^2}{\lambda^5} \frac{1}{e^{hc/\lambda kT} - 1}$$

Planck's formula

$$I = \sigma T^4$$

Stefan's law

$$\sigma = 5.67037 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4$$

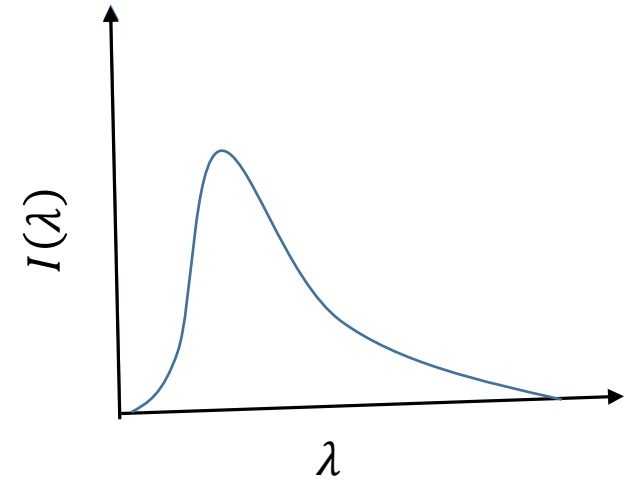
$$\lambda_{\text{max}} T = 2.8978 \times 10^{-3} \text{ m} \cdot \text{K}$$

Wien's displacement law

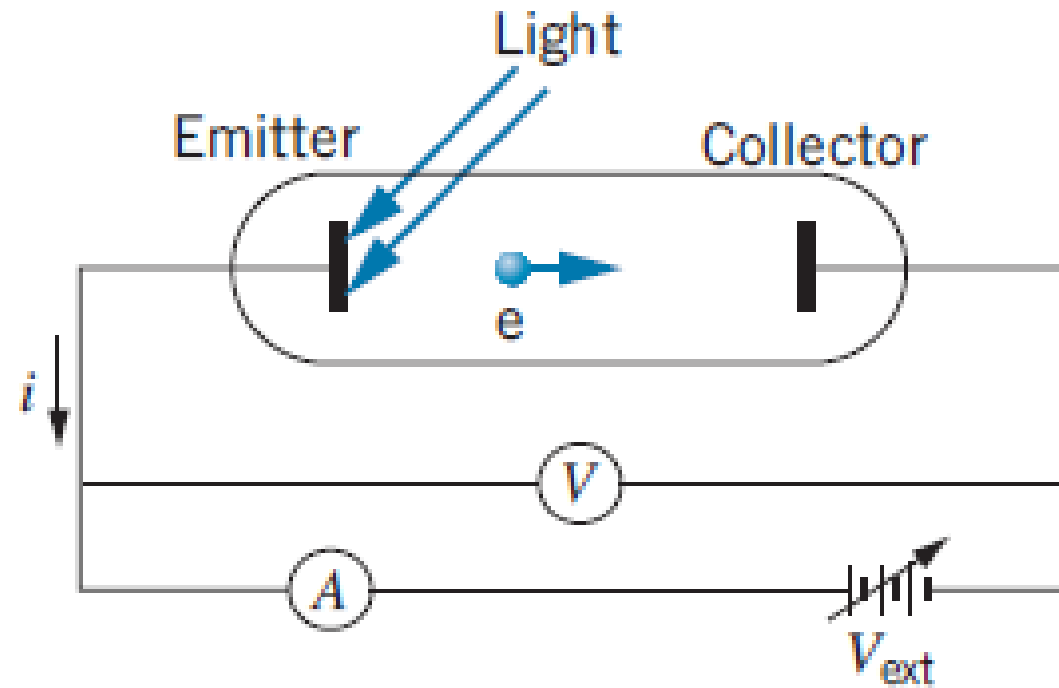
$$\sigma = \frac{2\pi^5 k^4}{15c^2 h^3}$$



$$h = 6.56 \times 10^{-34} \text{ J} \cdot \text{s}$$



Photoelectric effect



Photoelectric effect

The Classical Theory vs Experimental observation :

- **Theory:** The maximum kinetic energy of the electrons should be proportional to the intensity of the radiation.

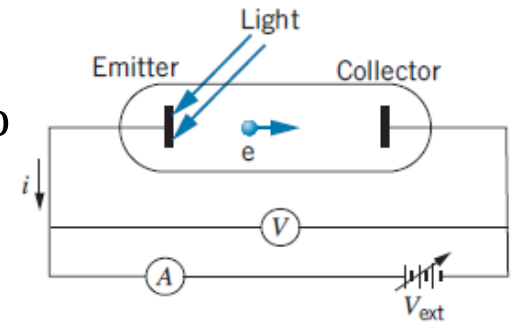
Observation : The maximum kinetic energy is not depend on the intensity but the frequency of the incident light

- **Theory:** The photoelectric effect should occur for light of any frequency or wavelength.

Observation : A minimum frequency of light is require to generate photoelectrons

- The first electrons should be emitted in a time interval of the order of seconds after the radiation begins to strike the surface.

Observation : Photo-emission is instantaneous !!!



Photoelectric effect

The Quantum Theory:

Einstein proposed that the energy of electromagnetic radiation is not continuously distributed over the wave front, but instead is concentrated in localized bundles or quanta (also known as photons)

The energy of a photon associated with an electromagnetic wave of frequency f is

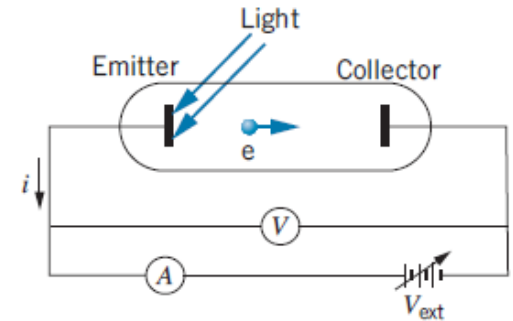
$$E = hf$$

Since photon is part of electromagnetic radiation, it is moving with speed 'c'

For a particle with finite mass m moves with 'c' possess infinite energy or momentum as

$$\gamma = \frac{1}{\sqrt{1 - \left(\frac{v}{c}\right)^2}} \rightarrow \infty \quad \text{So the mass of the photon should be zero}$$

$$E^2 = p^2 c^2 + \cancel{m^2 c^4} = p^2 c^2 \quad \Rightarrow \quad E = pc \quad p = \frac{E}{c} = \frac{h}{\lambda}$$



Photoelectric effect

The Quantum Theory:

Einstein proposed that the energy of electromagnetic radiation is not continuously distributed over the wave front, but instead is concentrated in localized bundles or quanta (also known as photons)

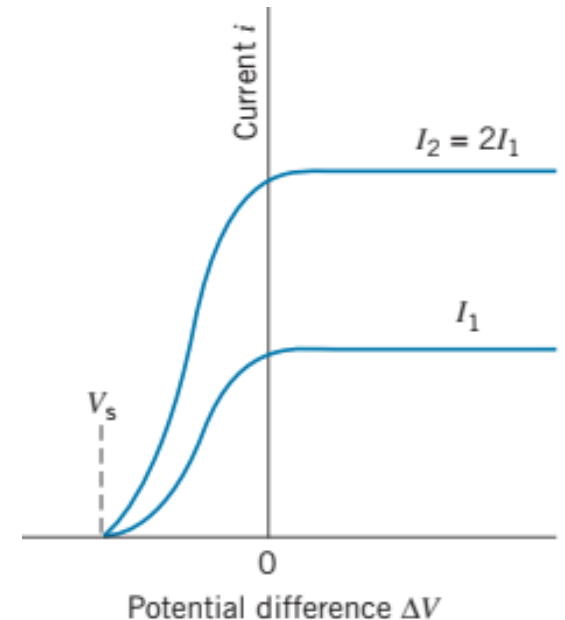
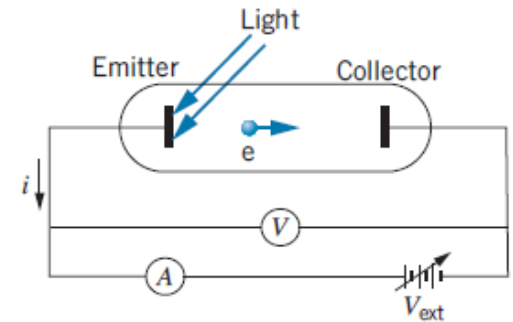
Interpretation: A photoelectron is released as a result of an encounter with a single photon. The entire energy of the photon is delivered instantaneously to a single photoelectron.

If the photon energy hf is greater than the work function ϕ of the material, the photoelectron will be released.

If the photon energy hf exceeds the work function, the excess energy appears as the kinetic energy of the electron:

$$K_{\max} = hf - \phi$$

(no role of intensity of the incident light !!!)



Photoelectric effect

The Quantum Theory:

$$K_{\max} = hf - \phi$$

For threshold frequency $K_{\max} = 0 \Rightarrow f_{th} = \phi/h$

If we apply the reverse potential then KE reduce.

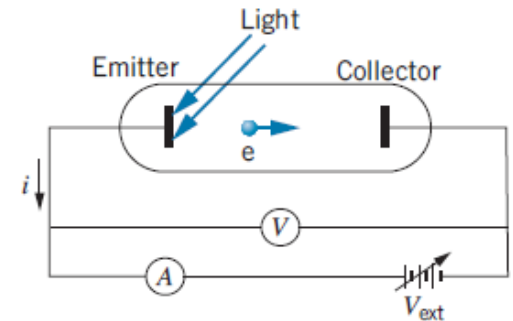
Minimum potential (V) require to stop the current is called the stopping potential.

In this situation potential gain by the photo electron (qV) is same as $K_{\max}$

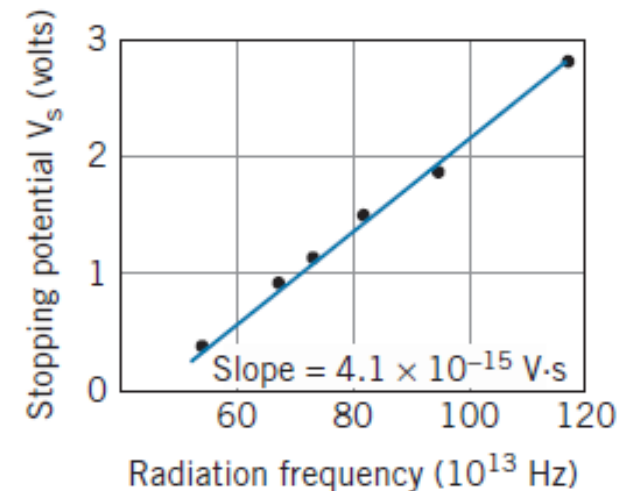
$$qV = hf - \phi$$

$$V = (h/q)f - (h/q)f_{th}$$

From the slope of the line, Millikan obtained a value for Planck's constant of $h = 6.57 \times 10^{-34} \text{ J} \cdot \text{s}$



Millikan's Exp.



Tutorials

Problem 1. The surface of the Sun has a temperature of about 6000 K. At what wavelength does the Sun emit its peak intensity? How does this compare with the peak sensitivity of the human eye?

Ans.

$$\lambda_{\max} T = 2.8978 \times 10^{-3} \text{ m} \cdot \text{K} \quad \text{Wien's displacement law}$$

$$\lambda = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{T} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{6000 \text{ K}} = 483 \text{ nm}$$

This is in the middle of the visible spectrum, close to the peak sensitivity of the eye.

Tutorials

Problem 2. The universe is filled with thermal radiation, which has a blackbody spectrum at an effective temperature of 2.7 K . What is the peak wavelength of this radiation? What is the energy (in eV) of quanta at the peak wavelength?

Ans.

$$\lambda_{\max} T = 2.8978 \times 10^{-3} \text{ m} \cdot \text{K} \quad \text{Wien's displacement law}$$

$$\lambda = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{T} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{2.7 \text{ K}} = 1.1 \text{ mm (microwave region)}$$

$$E = \frac{hc}{\lambda} = \frac{1240 \text{ eV} \cdot \text{nm}}{1.1 \text{ mm}} = 1.1 \times 10^{-3} \text{ eV}$$

Tutorials

Problem 3. (a) Assuming the human body (skin temperature 34°C) to behave like an ideal thermal radiator, find the wave length where the intensity from the body is a maximum. In what region of the electromagnetic spectrum is radiation with this wavelength? (b) Consider the height of a person is 6ft(1.83 m) and diameter 1ft(0.30 m) , estimate the power radiated by that person isolated from the surroundings. (c) Estimate the radiation power absorbed by a person in a room in which the temperature is 20°C

Ans.

$$\lambda = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{T} = \frac{2.898 \times 10^{-3} \text{ m} \cdot \text{K}}{307 \text{ K}} = 9.4 \mu\text{m (infrared)}$$

(b) Assume a person can be represented as a cylinder, about 6 feet (1.83 m) tall and 1 foot (0.30 m) in diameter. The surface area is $2\pi rL = 2\pi(0.15 \text{ m})(1.83 \text{ m}) = 1.72 \text{ m}^2$.

$$I = \sigma T^4 = (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(307 \text{ K})^4 = 504 \text{ W/m}^2$$

$$P = IA = (504 \text{ W/m}^2)(1.72 \text{ m}^2) = 870 \text{ W}$$

$$(c) \text{ For } T = 20^{\circ}\text{C} = 293 \text{ K} \quad I = \sigma T^4 = (5.67 \times 10^{-8} \text{ W/m}^2 \cdot \text{K}^4)(293 \text{ K})^4 = 418 \text{ W/m}^2$$

$$P = IA = (418 \text{ W/m}^2)(1.72 \text{ m}^2) = 719 \text{ W}$$

Thus the net power radiated by a person is about 150 W.

Tutorials

Problem 4. What is the cutoff wavelength for the photoelectric effect using an aluminum surface with work function 4.08 eV ?

Ans. $\phi = 4.08 \text{ eV}$

$$\lambda_c = \frac{hc}{\phi} = \frac{1240 \text{ eV} \cdot \text{nm}}{4.08 \text{ eV}} = 304 \text{ nm}$$

Tutorials

Problem 5. A metal surface has a photoelectric cutoff wavelength of 325.6 nm. It is illuminated with light of wavelength 259.8 nm. What is the stopping potential?

Ans.

$$\phi = \frac{hc}{\lambda_c} = \frac{1239.853 \text{ eV} \cdot \text{nm}}{325.6 \text{ nm}} = 3.808 \text{ eV}$$

$$eV_s = \frac{hc}{\lambda} - \phi = \frac{1239.853 \text{ eV} \cdot \text{nm}}{259.8 \text{ nm}} - 3.808 \text{ eV} = 0.964 \text{ eV}$$

$$V_s = 0.964 \text{ V}$$

Tutorials

Problem 6. A surface of zinc (with work function 4.31 eV) is illuminated and photoelectrons are observed. (a) What is the largest wavelength that will cause photoelectrons to be emitted? (b) What is the stopping potential when light of wavelength 220.0 nm is used?

Ans.

$$\lambda_c = \frac{hc}{\phi} = \frac{1240 \text{ eV} \cdot \text{nm}}{4.31 \text{ eV}} = 288 \text{ nm}$$

$$eV_s = \frac{hc}{\lambda} - \phi = \frac{1240 \text{ eV} \cdot \text{nm}}{220.0 \text{ nm}} - 4.31 \text{ eV} = 1.33 \text{ eV},$$

$$\text{so } V_s = 1.33 \text{ volts}$$

Photon absorption by a free electron

Let a free electron which is at rest absorbs a full photon incident on it with energy $h\nu$

The photon momentum is given by $E = pc$ as mass of the photon is zero

The momentum of the incident photon is $p = \frac{h\nu}{c}$ which is the total momentum before the collision

The total energy before the collision is $E_i = h\nu + m_0c^2$

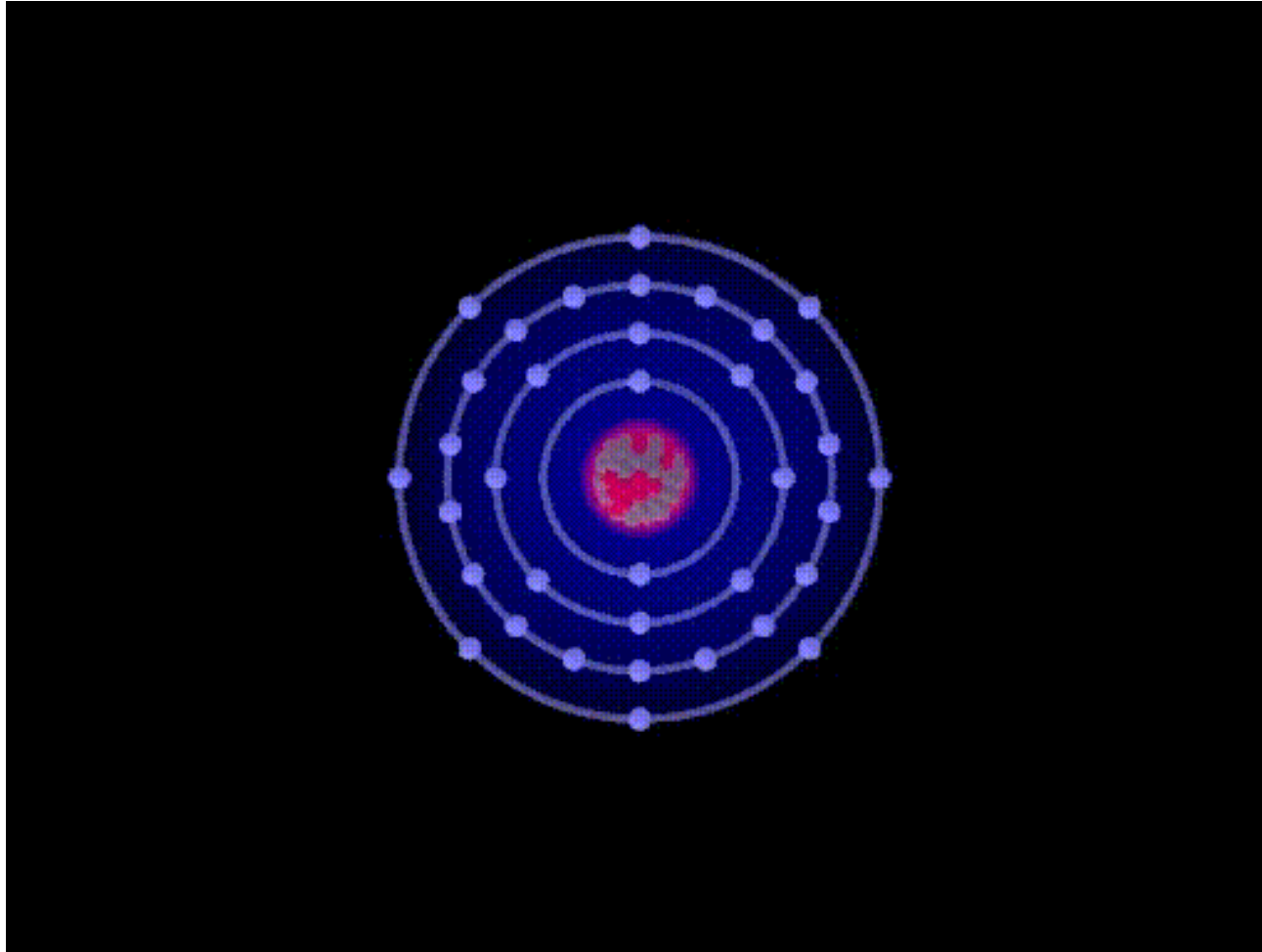
Since the photon is completely absorbed by the electron, from the conservation of momentum electron will have $p = \frac{h\nu}{c}$ momentum after the collision

The total energy after the collision is

$$E_f = \sqrt{(pc)^2 + (m_0c^2)^2} = \sqrt{(h\nu)^2 + (m_0c^2)^2} \neq E_i = h\nu + m_0c^2$$

THE COMPTON EFFECT

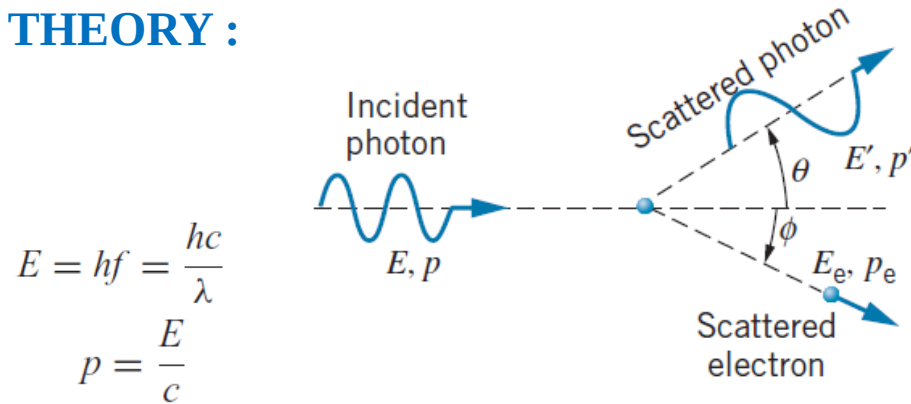
- Free electron can not absorb full energy of a photon
- If light is particle then the momentum will be conserved during the scattering process !!
- One can expect lower energy scattered photon



THE COMPTON EFFECT

- Free electron can not absorb full energy of a photon
- If light is particle then the momentum will be conserved during the scattering process !!
- One can expect lower energy scattered photon

THEORY :



$$E_{\text{initial}} = E_{\text{final}} : \quad E + m_e c^2 = E' + E_e$$

$$p_{x,\text{initial}} = p_{x,\text{final}} : \quad p = p_e \cos \phi + p' \cos \theta$$

$$p_{y,\text{initial}} = p_{y,\text{final}} : \quad 0 = p_e \sin \phi - p' \sin \theta$$

We have three equations with four unknowns θ, ϕ, E_e, E'

As p_e & p' can be obtain from E_e and E'

Eliminate variables associated with electron :

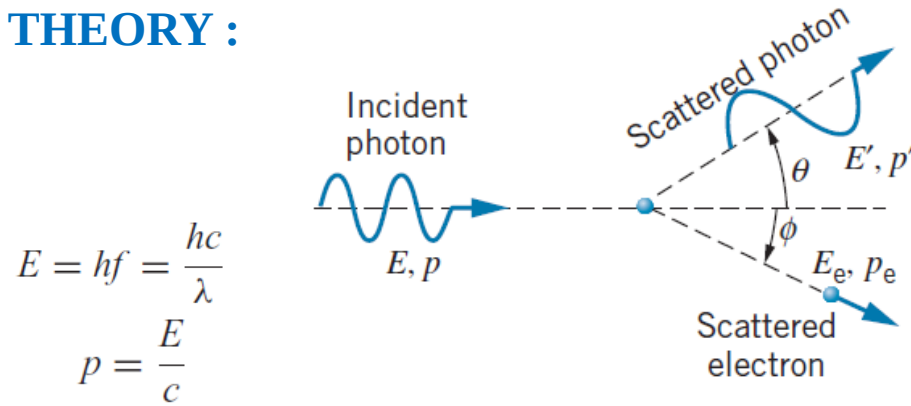
$$\left. \begin{aligned} p_e \cos \phi &= p - p' \cos \theta \\ p_e \sin \phi &= p' \sin \theta \end{aligned} \right\} p_e^2 = p^2 - 2pp' \cos \theta + p'^2$$

$$E_e^2 = c^2 p_e^2 + m_e^2 c^4 \quad \Rightarrow \quad (E + m_e c^2 - E')^2 = c^2 (p^2 - 2pp' \cos \theta + p'^2) + m_e^2 c^4$$

THE COMPTON EFFECT

- Free electron can not absorb full energy of a photon
- If light is particle then the momentum will be conserved during the scattering process !!
- One can expect lower energy scattered photon

THEORY :



$$(E + m_e c^2 - E')^2 = c^2(p^2 - 2pp' \cos \theta + p'^2) + m_e^2 c^4$$

$$\begin{aligned} \Rightarrow & \cancel{E^2} + \cancel{m_e^2 c^4} + \cancel{E'^2} + 2Em_e c^2 - 2EE' - 2E'm_e c^2 \\ & = \cancel{p^2 c^2} - 2pp' c^2 \cos \theta + \cancel{p'^2 c^2} + \cancel{m_e^2 c^4} \end{aligned}$$

$$\Rightarrow 2Em_e c^2 - 2EE' - 2E'm_e c^2 = -2pp' c^2 \cos \theta$$

$$\Rightarrow m_e c^2 (E - E') = EE' - pp' c^2 \cos \theta$$

$$\Rightarrow m_e c^2 (E - E') = EE' - EE' \cos \theta$$

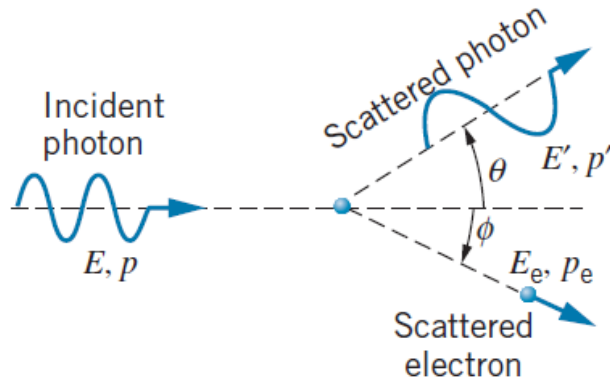
$$\Rightarrow m_e c^2 (E - E') = EE' (1 - \cos \theta)$$

$$\Rightarrow \frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2} (1 - \cos \theta)$$

THE COMPTON EFFECT

- Free electron can not absorb full energy of a photon
- If light is particle then the momentum will be conserved during the scattering process !!
- One can expect lower energy scattered photon

THEORY :



$$p = \frac{E}{c} = \frac{h\nu}{c} = \frac{hc}{c\lambda} = \frac{h}{\lambda}$$

$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2} (1 - \cos \theta)$$

$$\Rightarrow \frac{1}{p'c} - \frac{1}{pc} = \frac{1}{m_e c^2} (1 - \cos \theta)$$

$$\Rightarrow \frac{1}{p'} - \frac{1}{p} = \frac{1}{m_e c} (1 - \cos \theta)$$

$$\Rightarrow \lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$$

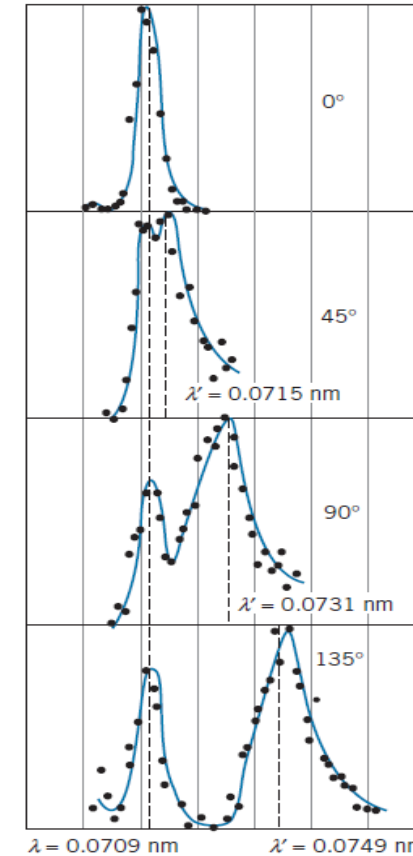
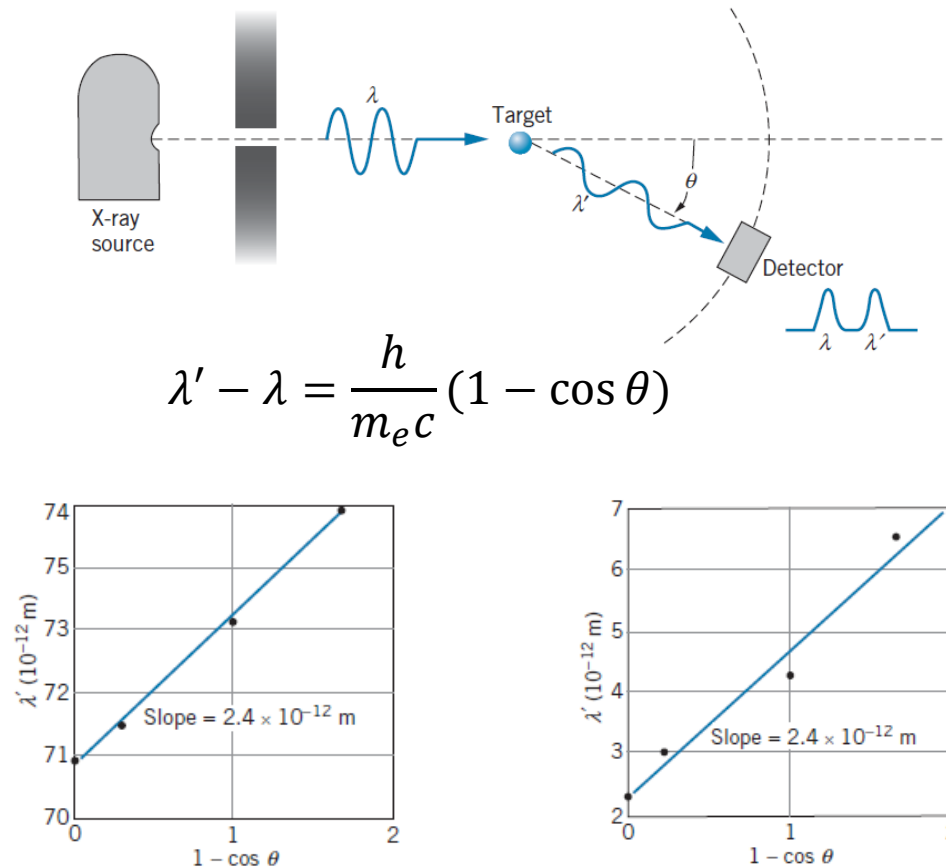
$$\lambda' \geq \lambda$$

$\frac{h}{m_e c}$ is called the *Compton wavelength of the electron* and has a value of 0.002426 nm

THE COMPTON EFFECT

- Free electron can not absorb full energy of a photon
- If light is particle then the momentum will be conserved during the scattering process !!
- One can expect lower energy scattered photon

Experiment :



THE COMPTON EFFECT

Electron gains energy :

$$E_e = K_e + m_e c^2,$$

$$E + m_e c^2 = E' + K_e + m_e c^2.$$

$$K_e = E - E'$$

The direction of the electron's motion :

$$p_e \cos \phi = p - p' \cos \theta$$

$$p_e \sin \phi = p' \sin \theta$$

$$\tan \phi = \frac{p_e \sin \phi}{p_e \cos \phi} = \frac{p' \sin \theta}{p - p' \cos \theta} = \frac{E' \sin \theta}{E - E' \cos \theta}$$

THE COMPTON EFFECT

Problem 1: Incident photons of energy 10.39 keV are Compton scattered, and the scattered beam is observed at 45.00° relative to the incident beam.

- (a) What is the energy of the scattered photons at that angle?
- (b) What is the kinetic energy of the scattered electrons?

Ans.: (a)
$$\frac{1}{E'} - \frac{1}{E} = \frac{1}{m_e c^2} (1 - \cos \theta)$$

$$\frac{1}{E'} = \frac{1}{E} + \frac{1 - \cos \theta}{m_e c^2} = \frac{1}{10.39 \text{ keV}} + \frac{1 - \sqrt{2}/2}{511.0 \text{ keV}} = 0.09682 \text{ keV}^{-1}$$

$$E' = 1 / 0.09682 \text{ keV}^{-1} = 10.33 \text{ keV} .$$

(b)
$$K_e = E_e - m_e c^2 = E - E' = 10.39 \text{ keV} - 10.33 \text{ keV} = 0.06 \text{ keV}$$

THE COMPTON EFFECT

Problem 2: X-ray photons of wavelength 0.02480 nm are incident on a target and the Compton-scattered photons are observed at 90.0°.

- (a) What is the wavelength of the scattered photons?
- (b) What is the momentum of the incident photons? Of the scattered photons?
- (c) What is the kinetic energy of the scattered electrons?
- (d) What is the momentum (magnitude and direction) of the scattered electrons?

Ans.: (a) $\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \theta)$

$$\lambda' = \lambda + (h / m_e c)(1 - \cos \theta) = 0.02480 \text{ nm} + (0.002426 \text{ nm})(1 - \cos 90^\circ) = 0.02723 \text{ nm}$$

(b) The momenta of the incident and scattered photons are:

$$p = \frac{h}{\lambda} = \frac{1}{c} \frac{hc}{\lambda} = \frac{1}{c} \frac{1240 \text{ eV} \cdot \text{nm}}{0.02480 \text{ nm}} = 4.999 \times 10^4 \text{ eV}/c \quad (x \text{ direction})$$

$$p' = \frac{h}{\lambda'} = \frac{1}{c} \frac{hc}{\lambda'} = \frac{1}{c} \frac{1240 \text{ eV} \cdot \text{nm}}{0.02723 \text{ nm}} = 4.553 \times 10^4 \text{ eV}/c \quad (y \text{ direction})$$

$$(c) \quad K_e = E - E' = cp - cp' = 4.999 \times 10^4 \text{ eV} - 4.553 \times 10^4 \text{ eV} = 4.46 \times 10^3 \text{ eV}$$

THE COMPTON EFFECT

Problem 2: X-ray photons of wavelength 0.02480 nm are incident on a target and the Compton-scattered photons are observed at 90.0°.

- (a) What is the wavelength of the scattered photons?
- (b) What is the momentum of the incident photons? Of the scattered photons?
- (c) What is the kinetic energy of the scattered electrons?
- (d) What is the momentum (magnitude and direction) of the scattered electrons?

Ans.: (d)

Because momentum must be conserved, the x component of the electron's momentum must equal p , and the y component must equal $-p'$:

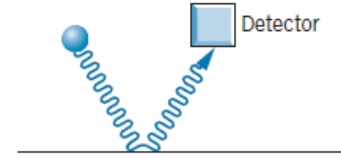
$$p_{\text{ex}} = p = 4.999 \times 10^4 \text{ eV}/c \quad \text{and} \quad p_{\text{ey}} = -p' = -4.553 \times 10^4 \text{ eV}/c$$

$$p_e = \sqrt{p_{\text{ex}}^2 + p_{\text{ey}}^2} = \sqrt{(4.999 \times 10^4 \text{ eV}/c)^2 + (4.553 \times 10^4 \text{ eV}/c)^2} = 6.762 \times 10^4 \text{ eV}/c$$

$$\text{in the direction given by } \theta = \tan^{-1} \frac{p_{\text{ey}}}{p_{\text{ex}}} = \tan^{-1} \frac{-4.553 \times 10^4 \text{ eV}/c}{4.999 \times 10^4 \text{ eV}/c} = -42.3^\circ$$

THE COMPTON EFFECT

Problem 3: High-energy gamma rays can reach a radiation detector by Compton scattering from the surroundings, as shown in the Figure. This effect is known as back-scattering. Show that, when $E \gg m_e c^2$, the back-scattered photon has an energy of approximately 0.25 MeV, independent of the energy of the original photon, when the scattering angle is nearly 180° .



Ans.:

$$\begin{aligned} \text{When } \theta = 180^\circ, \cos \theta = -1 \text{ and } \frac{1}{E'} &= \frac{1}{E} + \frac{2}{m_e c^2} \\ E' &= \frac{Em_e c^2}{2E + m_e c^2} \\ &= \frac{m_e c^2}{2 + m_e c^2 / E} \cong \frac{m_e c^2}{2} \quad \text{as } E \gg m_e c^2 \\ &= 0.25 \text{ MeV} \end{aligned}$$

THE COMPTON EFFECT

Problem 4: Gamma rays of energy 0.662MeV are Compton scattered.

- (a) What is the energy of the scattered photon observed at a scattering angle of 60.0°?
- (b) What is the kinetic energy of the scattered electrons?

Ans.:

$$(a) \quad \frac{1}{E'} = \frac{1}{E} + \frac{1 - \cos \theta}{m_e c^2} = \frac{1}{0.662 \text{ MeV}} + \frac{1 - \cos 60^\circ}{0.511 \text{ MeV}} = 2.489 \text{ MeV}^{-1}$$

$$\text{so } E' = 1 / 2.489 \text{ MeV}^{-1} = 0.402 \text{ MeV}.$$

$$(b) \quad K_e = E - E' = 0.662 \text{ MeV} - 0.402 \text{ MeV} = 0.260 \text{ MeV}$$

Wave nature of particle

WAVE NATURE OF PARTICLES

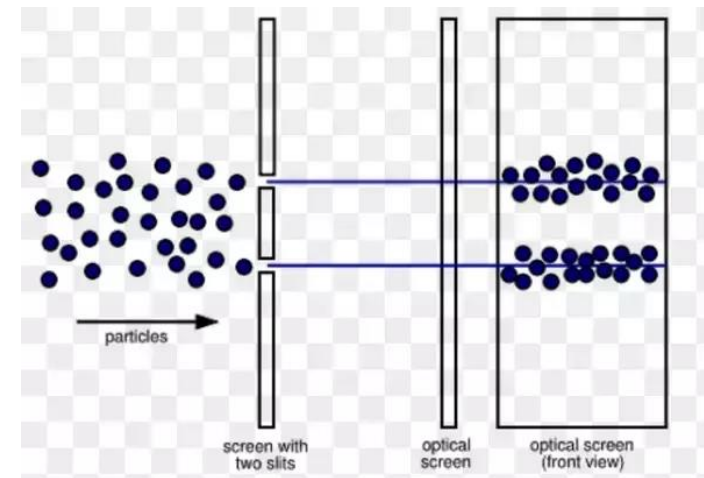
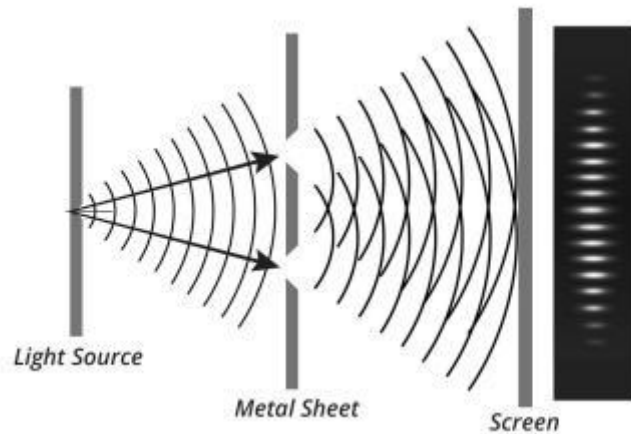
In case of photon $E = hf$, and $p = h/\lambda$

DE BROGLIE'S HYPOTHESIS :

Any material particle moving with momentum p , there is a wave of wavelength λ associated with it such that

$$\lambda = \frac{h}{p}$$

Double slit experiment :



WAVE NATURE OF PARTICLES

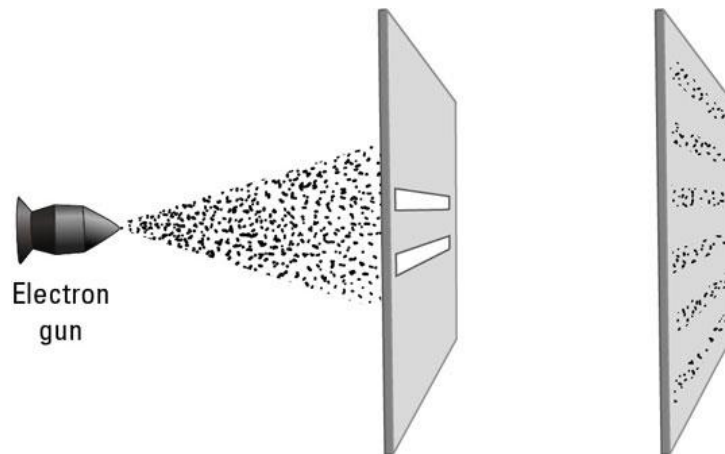
In case of photon $E = hf$, and $p = h/\lambda$

DE BROGLIE'S HYPOTHESIS :

Any material particle moving with momentum p , there is a wave of wavelength λ associated with it such that

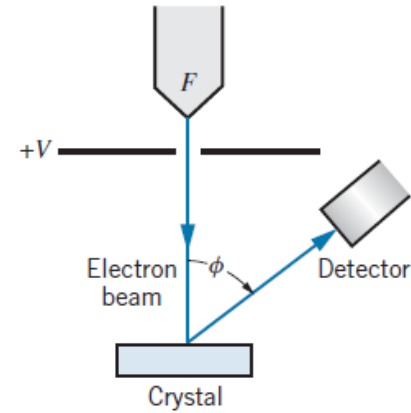
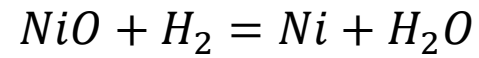
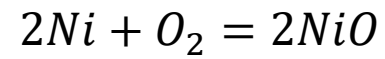
$$\lambda = \frac{h}{p}$$

Double slit experiment :



Davisson and Germer Experiment

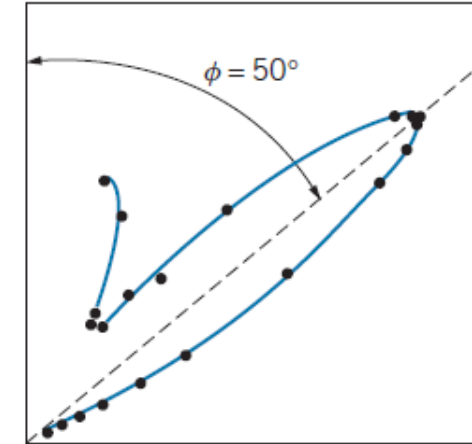
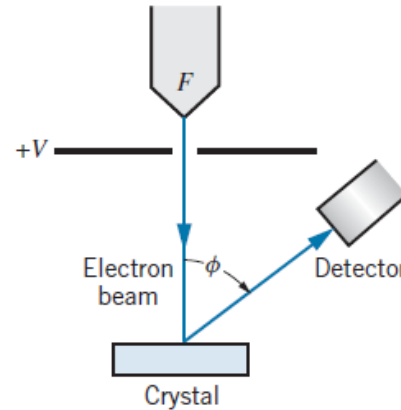
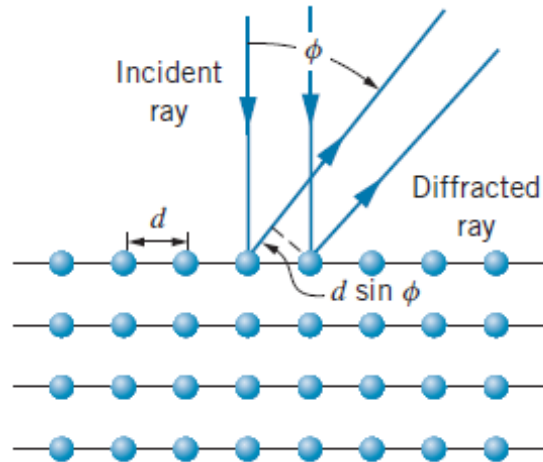
Electron scattering from polycrystalline Ni



Davisson and Germer Experiment

When the accelerating voltage is set at 54 V, there is an intense reflection of the beam at the angle $\phi = 50^\circ$.

Diffraction of Wave Theory



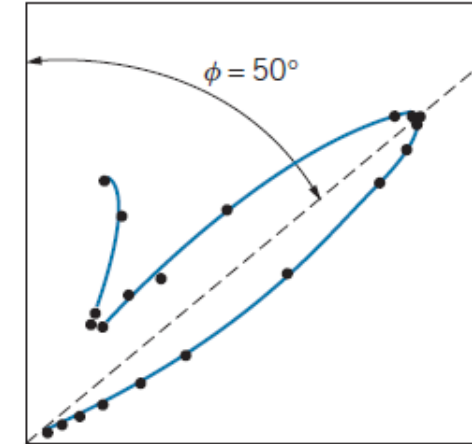
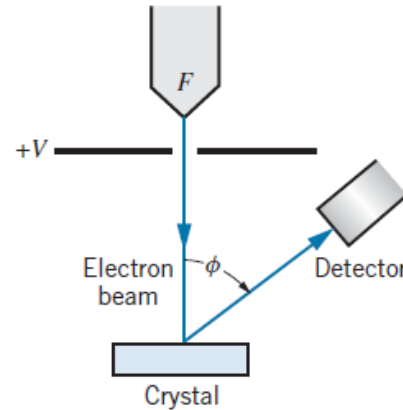
$$d \sin \phi = n\lambda \quad n = 1, 2, 3, \dots$$

$$\lambda = d \sin \phi = (0.215 \text{ nm})(\sin 50^\circ) = 0.165 \text{ nm}$$

Davisson and Germer Experiment

When the accelerating voltage is set at 54 V, there is an intense reflection of the beam at the angle $\phi = 50^\circ$.

Possible explanation of result from de Broglie Theory



The electron accelerated through a potential difference of 54 V has a kinetic energy of 54 eV. Therefore, momentum of the electron was

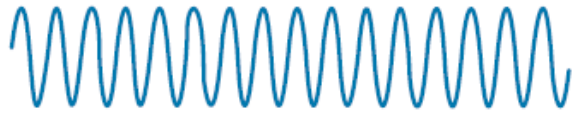
$$p = \sqrt{2mK} = \frac{1}{c}\sqrt{2mc^2K} = \frac{1}{c}\sqrt{2(511,000 \text{ eV})(54 \text{ eV})} = \frac{1}{c}(7430 \text{ eV})$$

The de Broglie wavelength is $\lambda = h/p = hc/pc$. $hc = 1240 \text{ eV} \cdot \text{nm}$,

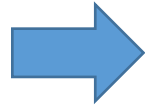
$$\lambda = \frac{hc}{pc} = \frac{1240 \text{ eV} \cdot \text{nm}}{7430 \text{ eV}} = 0.167 \text{ nm}$$

UNCERTAINTY

If we want to use de Broglie waves to describe particles, the amplitude of the wave will specify the location of the particle.



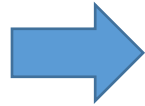
Sinusoidal wave



wave extends from $-\infty$ to $+\infty$, so the particle might be found anywhere in that region.



wave pulse



Locating the particle in a small region of space

Simultaneous measurement of position and momentum of a particle produces error

UNCERTAINTY

Heisenberg uncertainty principle:

- It is not possible to make a simultaneous determination of the position and the momentum of a particle with unlimited precision,
- It is not possible to make a simultaneous determination of the energy and the time coordinate of a particle with unlimited precision.

Heisenberg uncertainty relation :

$$\Delta x \Delta p_x \geq \frac{1}{2} \hbar \quad \Rightarrow \quad \Delta x \Delta p_x \sim \hbar$$

$$\Delta E \Delta t \geq \frac{1}{2} \hbar \quad \Rightarrow \quad \Delta E \Delta t \sim \hbar$$

UNCERTAINTY

Example: An electron moves in the x direction with a speed of 3.6×10^6 m/s. We can measure its speed to a precision of 1%. With what precision can we simultaneously measure its x coordinate?

Repeat the calculations for the case of a pitched baseball ($m = 0.145$ kg) moving at a speed of 42.5 m/s. Again assume that its speed can be measured to a precision of 1%.

Ans. The electron's momentum is
$$p_x = mv_x = (9.11 \times 10^{-31} \text{ kg})(3.6 \times 10^6 \text{ m/s})$$
$$= 3.3 \times 10^{-24} \text{ kg} \cdot \text{m/s}$$

The uncertainty Δp_x is 1% of this value, or 3.3×10^{-26} kg · m/s.

The uncertainty in position is then
$$\Delta x \sim \frac{\hbar}{\Delta p_x} = \frac{1.05 \times 10^{-34} \text{ J} \cdot \text{s}}{3.3 \times 10^{-26} \text{ kg} \cdot \text{m/s}} = 3.2 \text{ nm}$$

In case of baseball

The baseball's momentum is
$$p_x = mv_x = (0.145 \text{ kg})(42.5 \text{ m/s}) = 6.16 \text{ kg} \cdot \text{m/s}$$

The uncertainty in momentum is 6.16×10^{-2} kg · m/s,

The uncertainty in position is then
$$\Delta x \sim \frac{\hbar}{\Delta p_x} = \frac{1.05 \times 10^{-34} \text{ J} \cdot \text{s}}{6.16 \times 10^{-2} \text{ kg} \cdot \text{m/s}} = 1.7 \times 10^{-33} \text{ m}$$

UNCERTAINTY

Example: (a) A charged pi meson has a rest energy of 140MeV and a lifetime of 26 ns. Find the energy uncertainty of the pi meson, expressed in MeV and also as a fraction of its rest energy.

(b) Repeat for the uncharged pi meson, with a rest energy of 135MeV and a lifetime of 8.3×10^{-17} s.

(c) Repeat for the rho meson, with a rest energy of 765MeV and a lifetime of 4.4×10^{-24} s.

Ans. (a) The pi meson lives for 26 ns, we have only that much time to measure its rest energy.

From uncertainty relation we can say that, any energy measurement done in a Δt interval is uncertain by an amount of at least

$$\Delta E = \frac{\hbar}{\Delta t} = \frac{6.58 \times 10^{-16} \text{ eV} \cdot \text{s}}{26 \times 10^{-9} \text{ s}} = 2.5 \times 10^{-8} \text{ eV}$$

$$\frac{\Delta E}{E} = \frac{2.5 \times 10^{-14} \text{ MeV}}{140 \text{ MeV}} = 1.8 \times 10^{-16}$$

(b) For uncharged pi-meson:
$$\Delta E = \frac{\hbar}{\Delta t} = \frac{6.58 \times 10^{-16} \text{ eV} \cdot \text{s}}{8.3 \times 10^{-17} \text{ s}} = 7.9 \text{ eV} = 7.9 \times 10^{-6} \text{ MeV}$$

$$\frac{\Delta E}{E} = \frac{7.9 \times 10^{-6} \text{ MeV}}{135 \text{ MeV}} = 5.9 \times 10^{-8}$$

(c) For uncharged rho-meson:
$$\Delta E = \frac{\hbar}{\Delta t} = \frac{6.58 \times 10^{-16} \text{ eV} \cdot \text{s}}{4.4 \times 10^{-24} \text{ s}} = 1.5 \times 10^8 \text{ eV} = 150 \text{ MeV}$$

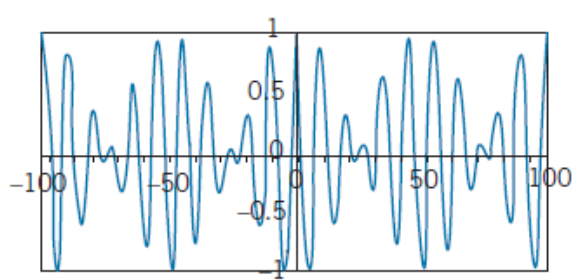
$$\frac{\Delta E}{E} = \frac{150 \text{ MeV}}{765 \text{ MeV}} = 0.20$$

WAVE PACKETS

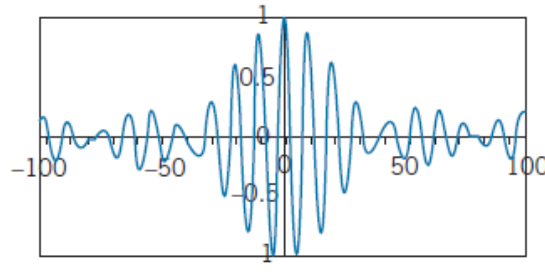
Associating a wave with single wave length will not be suitable to describe a particle as position of the particle will be completely undermined.

We may use group of wave or wave packet instead

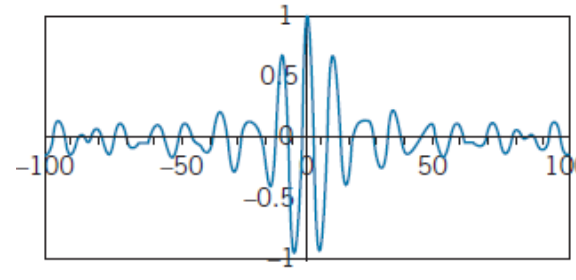
$$y(x) = A_1 \cos k_1 x + A_2 \cos k_2 x = A_1 \cos(2\pi x/\lambda_1) + A_2 \cos(2\pi x/\lambda_2)$$



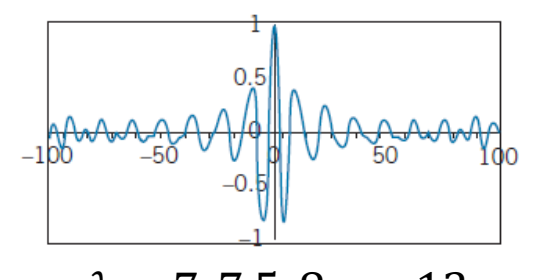
$$\lambda_1 = 9, \lambda_2 = 11$$



$$\lambda = 9, 9.5, 10, 10.5, 11$$

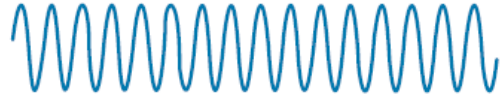


$$\lambda = 8, 8.5, 9, \dots, 12$$

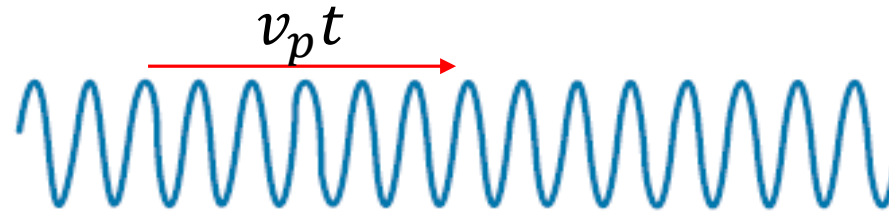


$$\lambda = 7, 7.5, 8, \dots, 13$$

WAVE PACKETS

Wave solution: $y(x) = A \cos(kx)$ 

Travelling wave : Same phase of the wave moving with a velocity v_p



If at time $t=0$, $x=0$ has the maximum amplitude, then this maximum amplitude will move to a distance $v_p t$ at time t .

We therefore need to replace x by $(x - v_p t)$ in the wave equation $y(x) = A \cos(kx)$

Travelling wave equation :
$$\begin{aligned} y(x, t) &= A \cos[k(x - v_p t)] \\ &= A \cos(kx - kv_p t) \\ &= A \cos(kx - \omega t) \end{aligned}$$

$$\begin{aligned} kv_p &= \frac{2\pi}{\lambda} \times \lambda f \\ &= 2\pi f \\ &= \omega \end{aligned}$$

WAVE PACKETS

Let us consider two travelling wave

$$y_1(x, t) = A \cos(k_1 x - \omega_1 t) \quad \& \quad y_2(x, t) = A \cos(k_2 x - \omega_2 t)$$

$$y(x, t) = y_1(x, t) + y_2(x, t) = A [\cos(k_1 x - \omega_1 t) + \cos(k_2 x - \omega_2 t)]$$

$$= 2A \cos\left(\frac{k_1 - k_2}{2}x - \frac{\omega_1 - \omega_2}{2}t\right) \cos\left(\frac{k_1 + k_2}{2}x - \frac{\omega_1 + \omega_2}{2}t\right)$$

let $k_1 = k$ and $k_2 = k + \Delta k$

$$y(x, t) = 2A \cos\left(\frac{\Delta k}{2}x - \frac{\Delta \omega}{2}t\right) \cos(kx - \omega t)$$

WAVE PACKETS

$$y(x, t) = 2A \cos\left(\frac{\Delta k}{2}x - \frac{\Delta \omega}{2}t\right) \cos(kx - \omega t)$$

There is a modulation of amplitude of the resultant wave with time !!!

Modulation also has motion

$$\text{Modulation velocity } v = \frac{\Delta \omega}{\Delta k}$$

If we consider many wave with continuous change in wave number and frequency

$$v = \frac{d\omega}{dk}$$

WAVE PACKETS

$$v = \frac{d\omega}{dk}$$



Velocity of the resultant amplitude for a group of wave, called group velocity v_g

$$\omega = 2\pi f = \frac{2\pi E}{h} = \frac{2\pi(\gamma mc^2)}{h} = \frac{2\pi mc^2}{h} \sqrt{\frac{1}{1 - \frac{v^2}{c^2}}}$$

$$k = \frac{2\pi}{\lambda} = \frac{2\pi p}{h} = \frac{2\pi \gamma mv}{h} = \frac{2\pi mv}{h} \sqrt{\frac{1}{1 - \frac{v^2}{c^2}}}$$

$$v_g = \frac{d\omega}{dk} = v$$

$$v_p = \frac{\omega}{k}$$

$$v_g = \frac{d\omega}{dk} = \frac{d\omega/dv}{dk/dv}$$

$$= \frac{\frac{2\pi mv}{h \left(1 - \frac{v^2}{c^2}\right)^{3/2}}}{\frac{2\pi m}{h \left(1 - \frac{v^2}{c^2}\right)^{3/2}}} = v$$

WAVE PACKETS

Example 1 : Find the group velocity and the phase velocity of an electron with KE 500 KeV

Ans. :

$$KE = (\gamma - 1)mc^2 \quad \longrightarrow \quad v = 0.8626c = v_g$$

$$v_p = \frac{c^2}{v} = 1.1592 c$$

WAVE PACKETS

Example 2 : The phase velocity of ocean wave is $\sqrt{\frac{g\lambda}{2\pi}}$, where g is the gravitational constant. Find the group velocity of ocean wave

WAVE FUNCTION & SCHRODINGER EQUATION

- ❖ States in quantum mechanics is not pure due to uncertainty.
- ❖ States are determined probabilistically
- ❖ Initial state can not determine the future states !!
- ❖ Particle are represented by wave (probability wave)
- ❖ Mathematical form of wave is the wave function $\Psi(r, t)$
- ❖ In general wave functions are complex function
- ❖ $\Psi(r, t) * \Psi(r, t) = |\Psi(r, t)|^2$ is the probability of finding the particle
- ❖ $\int_{-\infty}^{\infty} \Psi(r, t) * \Psi(r, t) dr = 1$

WAVE FUNCTION & SCHRODINGER EQUATION

Well behaved wavefunction :

- ❖ $\Psi(r, t)$ Must be single valued and continuous everywhere in space
- ❖ $\frac{\partial \Psi(r, t)}{\partial x}$ Must be continuous and single valued
- ❖ $\Psi(r, t)$ must be normalizable. i.e. $\Psi(r, t) \rightarrow 0$ as $x \rightarrow \pm\infty$
- ❖ For a given such wave function, the probability of finding a particle in the interval (x_1, x_2) as

$$\int_{x_1}^{x_2} \Psi(r, t) * \Psi(r, t) dr = 1$$

WAVE FUNCTION & SCHRODINGER EQUATION

Wave-equation :

$$\frac{\partial^2 Y(x, t)}{\partial x^2} = \frac{1}{v^2} \frac{\partial^2 Y(x, t)}{\partial t^2}$$

The solution should be of the form $Y(x, t) = f(x \pm vt)$

The general real solution : $Y(x, t) = A \cos \left[\omega \left(t - \frac{x}{v} \right) \right]$ or $= A \sin \left[\omega \left(t - \frac{x}{v} \right) \right]$

Most general complex solution : $Y(x, t) = A \exp \left[-i\omega \left(t - \frac{x}{v} \right) \right]$

Associate this wave function with a free particle
with energy E and momentum p :

WAVE FUNCTION & SCHRÖDINGER EQUATION

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